

STATE TECHNICAL UNIVERSITY

END SEMESTER EXAMINATION 2025-2026 | ANSWER BOOKLET

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EXAMINATION / COURSE: B.Tech Semester II
SUBJECT & CODE: Physics (CS 201)
BOOKLET SERIAL NO: BK-PH-80047

BARCODE SCAN TRACKING



BK-PH-80047

Q.No	Q1	Q2	Q3	Q4	Q5
Max	14	14	14	14	14
Marks					

START WRITING YOUR ANSWERS BELOW. DO NOT WRITE IN THE MARGINS.

Q.1 Ans 1. Quantum Physics: Group Velocity & Particle Velocity:

In quantum mechanics, a moving physical particle (mass m , momentum p) is associated with a wave packet rather than a single harmonic wave (de Broglie hypothesis: $\lambda = h/p$).

1. Phase Velocity (v_p):

Speed of propagation of individual phase crests of a monochromatic wave:

$$v_p = \omega / k = E / p = (p^2 / 2m) / p = p / (2m) = v / 2 \text{ (non-relativistic).}$$

2. Group Velocity (v_g):

Speed with which the overall modulation envelope (wave packet) moves:

$$v_g = d(\omega) / dk = dE / dp.$$

$$\text{Since } E = p^2 / (2m), dE/dp = 2p / (2m) = p / m = v \text{ (particle velocity).}$$

$$\text{Hence, } v_g = v_{\text{particle}}.$$

3. Relativistic Proof:

$$E^2 = p^2 c^2 + m^2 c^4.$$

$$\text{Differentiating: } 2E dE = 2p c^2 dp \Rightarrow dE/dp = (p c^2) / E.$$

$$\text{Substituting } p = \gamma m_0 v \text{ and } E = \gamma m_0 c^2:$$

$$v_g = (\gamma m_0 v c^2) / (\gamma m_0 c^2) = v.$$

Therefore, group velocity v_g is universally equal to classical particle velocity v .

Q.2

Ans 2. Wave Optics - Interference of Light:

Interference is the redistribution of luminous energy due to superposition of two coherent wave trains:

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\Delta\phi).$$

Conditions for Sustained Interference:

1. Sources must be strictly coherent (constant phase difference).
2. Sources must be monochromatic (single wavelength λ).
3. Equal or nearly equal amplitudes for maximum fringe contrast.

Division of Wavefront vs Division of Amplitude:

a) Division of Wavefront:

- Original wavefront is divided into two parts by mirrors, prisms or slits.
- Examples: Young's Double Slit, Fresnel Biprism, Lloyd's Mirror.
- Requires a point or narrow slit source. Fringe width $\beta = \lambda D / d$.

b) Division of Amplitude:

- Incident wave amplitude is split by partial reflection and refraction at interfaces.
- Examples: Thin dielectric films, Newton's Rings, Michelson Interferometer.
- Can use broad extended sources, producing high intensity fringes.
- Path difference in thin film: $\Delta = 2 \mu t \cos(r) - \lambda / 2$ (Stokes' phase shift).
- Applications: Anti-reflection optical coatings ($t = \lambda / 4\mu$) and optical surface testing.

Q.3

Ans 3. Lasers and Fibre Optics:

1. Laser Principles (Light Amplification by Stimulated Emission of Radiation):

- Stimulated Absorption: Ground atom absorbs $h\nu$ and jumps to E_2 .
- Spontaneous Emission: Random downward transition emitting photon: $\text{Rate} = A_{21} * N_2$.
- Stimulated Emission: Incident photon induces excited atom to emit an identical twin photon (same frequency, phase, direction): $\text{Rate} = B_{21} * N_2 * \rho(\nu)$.
- Population Inversion ($N_2 > N_1$): Non-equilibrium state achieved via optical/electrical pumping in a 3-level or 4-level system with a metastable state (lifetime $\sim 10^{-3}$ s).
- Optical Resonator: Pair of mirrors (100% and 95% reflection) creating standing waves: $L = m * \lambda / 2$.

2. Fibre Optics Fundamentals:

- Wave propagation occurs by Total Internal Reflection (TIR) at Core-Cladding interface ($n_1 > n_2$).
- Critical Angle: $\theta_c = \sin^{-1}(n_2 / n_1)$.
- Acceptance Angle (θ_o): Maximum entry angle:
 $\sin(\theta_o) = \sqrt{n_1^2 - n_2^2} = \text{NA}$ (Numerical Aperture).
- Fractional Refractive Index Change: $\Delta = (n_1 - n_2) / n_1$.
- $\text{NA} = n_1 * \sqrt{2 * \Delta}$.
- Types: Step Index (Single mode / Multimode) and Graded Index (GRIN) fibers with parabolic index profile reducing intermodal dispersion.

Q.4

Ans 4. Electromagnetism - Maxwell's Equations & Boundary Conditions:

1. Maxwell's Four Fundamental Equations:

a) Gauss's Law for Electrostatics:

$$\text{Differential: } \text{div}(\mathbf{D}) = \rho_v \quad | \quad \text{Integral: } \oiint \mathbf{D} \cdot d\mathbf{A} = Q_{\text{enclosed}}$$

Physical meaning: Electric flux through closed surface equals total enclosed charge.

b) Gauss's Law for Magnetism:

$$\text{Differential: } \text{div}(\mathbf{B}) = 0 \quad | \quad \text{Integral: } \oiint \mathbf{B} \cdot d\mathbf{A} = 0$$

Physical meaning: Magnetic monopoles do not exist; magnetic lines form continuous loops.

c) Faraday's Law of Electromagnetic Induction:

$$\text{Differential: } \text{curl}(\mathbf{E}) = -d\mathbf{B}/dt \quad | \quad \text{Integral: } \oint \mathbf{E} \cdot d\mathbf{l} = -d(\Phi_B)/dt$$

Physical meaning: Time-varying magnetic fields induce circulating electric fields.

d) Ampere-Maxwell Law:

$$\text{Differential: } \text{curl}(\mathbf{H}) = \mathbf{J} + d\mathbf{D}/dt \quad | \quad \text{Integral: } \oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{cond}} + d(\Phi_E)/dt$$

Displacement current density: $\mathbf{J}_d = d\mathbf{D}/dt = \epsilon \cdot d\mathbf{E}/dt$, resolving continuity equation $\text{div}(\mathbf{J}) + d\rho/dt = 0$.

2. Boundary Conditions at Dielectric Interfaces:

- Tangential $\mathbf{E}$ -field is continuous: $\mathbf{E}_{1_t} = \mathbf{E}_{2_t}$.
- Normal $\mathbf{D}$ -field discontinuity: $\mathbf{D}_{1_n} - \mathbf{D}_{2_n} = \rho_s$ (if $\rho_s = 0$, $\mathbf{D}_{1_n} = \mathbf{D}_{2_n}$).
- Tangential $\mathbf{H}$ -field continuity (for non-conducting boundary): $\mathbf{H}_{1_t} = \mathbf{H}_{2_t}$.
- Normal $\mathbf{B}$ -field is always continuous: $\mathbf{B}_{1_n} = \mathbf{B}_{2_n}$.

3. Poynting Vector: $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ (W/m^2), representing instantaneous directional power flow per unit area.

Q.5

Ans 5. Solid State & Semiconductor Physics:

1. Crystal Structures and Lattice Geometry:

- Simple Cubic (SC): Atoms at 8 corners, Coordination No (CN) = 6, Atomic Packing Factor (APF) = 52%.
- Body Centered Cubic (BCC): Atoms at corners + body center, CN = 8, APF = 68%.
- Face Centered Cubic (FCC): Atoms at corners + face centers, CN = 12, APF = 74%.
- Miller Indices (h k l): Reciprocal of planar intercepts cleared of fractions. Interplanar distance: $d = a / \sqrt{h^2 + k^2 + l^2}$.

2. Energy Band Theory:

- Kronig-Penney Model explains periodic lattice potential causing splitting of energy levels into allowed bands and forbidden gaps (E_g).
- Conductors ($E_g = 0$ eV), Semiconductors ($E_g \sim 1.1$ eV for Si), Insulators ($E_g > 5$ eV).

3. Intrinsic vs Extrinsic Semiconductors:

- Intrinsic (pure Si): $n_i = p_i = \sqrt{N_c \cdot N_v} \cdot \exp(-E_g / 2kT)$. Fermi level E_f lies in middle of bandgap.
- n-type: Doped with pentavalent donors (P, As); majority carriers electrons; E_f shifts close to conduction band.
- p-type: Doped with trivalent acceptors (B, Al); majority carriers holes; E_f shifts close to valence band.

4. Direct vs Indirect Bandgap:

- Direct (e.g. GaAs): Conduction band minimum & valence band maximum align at $k = 0$ ($\Delta k = 0$); efficient radiative recombination (ideal for LEDs/lasers).
- Indirect (e.g. Si): Peak and valley are misaligned; requires phonon participation, releasing heat.

--- End of Answer Booklet (All 5 questions attempted) ---

Candidate Sign: Priya